

Web Programming Course
Exam Simulation – May 28, 2025

- This is an open-book exam, but all work must be done individually.
- Answer each question clearly, using sketches, tables, or descriptions where necessary.
- Provide examples of requests and responses to demonstrate functionality.
- Do not write actual code, but provide detailed API designs, request/response formats, and data models.

Design a web system that manages reservations in a restaurant. The system must include:

- An API to register a new restaurant (name, address, maximum number of seats);
- An API to retrieve the list of all available restaurants;
- An API to view the details of a restaurant given its ID;
- An API to book a table by providing name, number of people, and time.

Question 1: API Definition (15 points)

- Define API endpoints (URL + HTTP method) and describe them textually where necessary – 2 points for each API
- Specify request parameters and expected responses – 1 point for each API
- Define the data models for request and response – 3 points total

Question 2: Error Handling (5 points)

- Describe relevant error codes (400, 404, 500) – 3 points
- Provide at least two concrete examples of error responses – 1 point each

Some possible 3-point questions:

1. Suppose the system is used simultaneously by thousands of users during peak hours (e.g., Saturday dinner). List at least two technical strategies you could apply to improve scalability and performance of the system. Briefly explain the role of each.
2. Describe two best practices for the automatic deployment of the restaurant reservation system so that it can be easily updated and remain reliable in production.
3. The current system allows anyone to make reservations or register restaurants. Explain how to introduce user authentication and manage different roles (e.g., regular user vs. restaurant manager).
4. After a user makes a reservation, the system must send them a confirmation email. Explain why it's better to handle email sending asynchronously, and which tools you could use to implement it.
5. To ensure that the APIs work correctly, it's important to test them. Describe at least two types of automated tests you would write for this project and which part of the system they would verify.